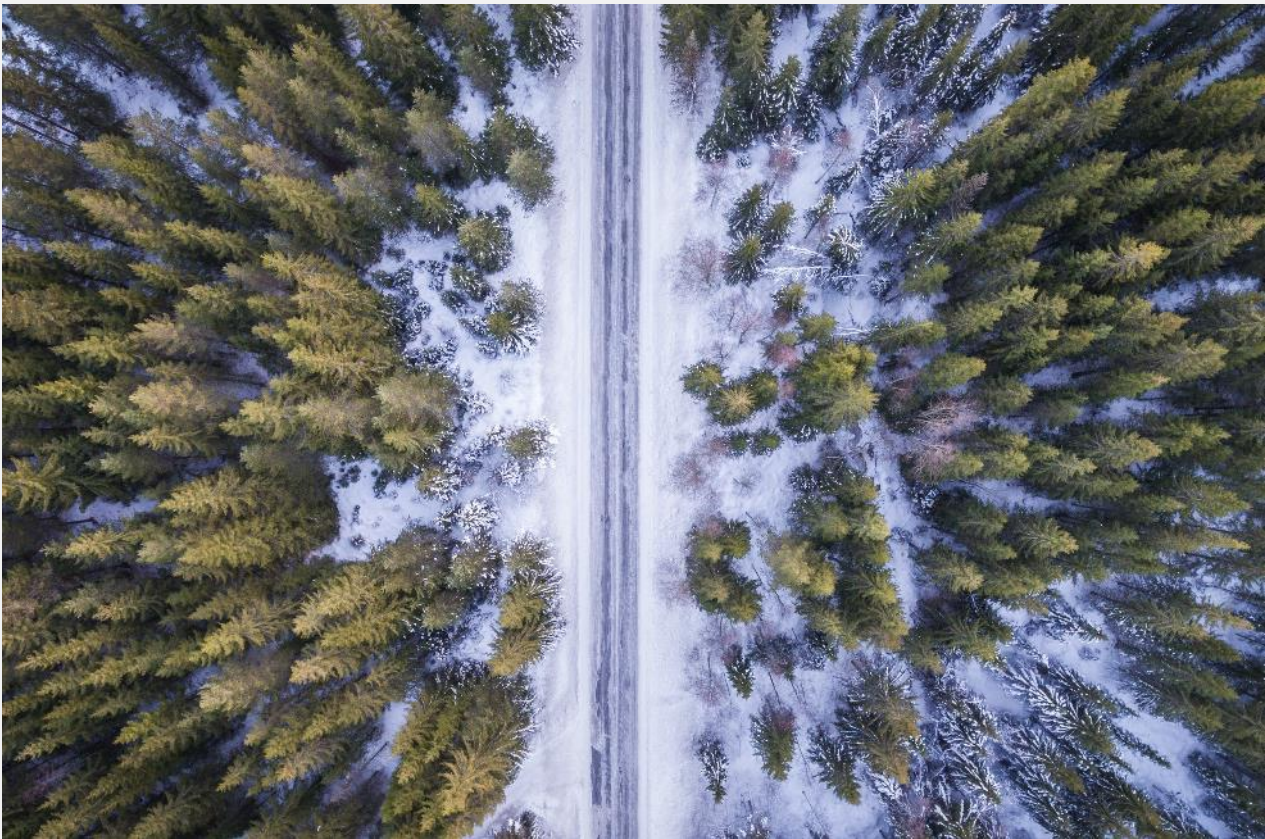


Mobile-Based Convolutional Neural Network Approach for Real-Time Maize Leaf Disease Detection



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Abarna P



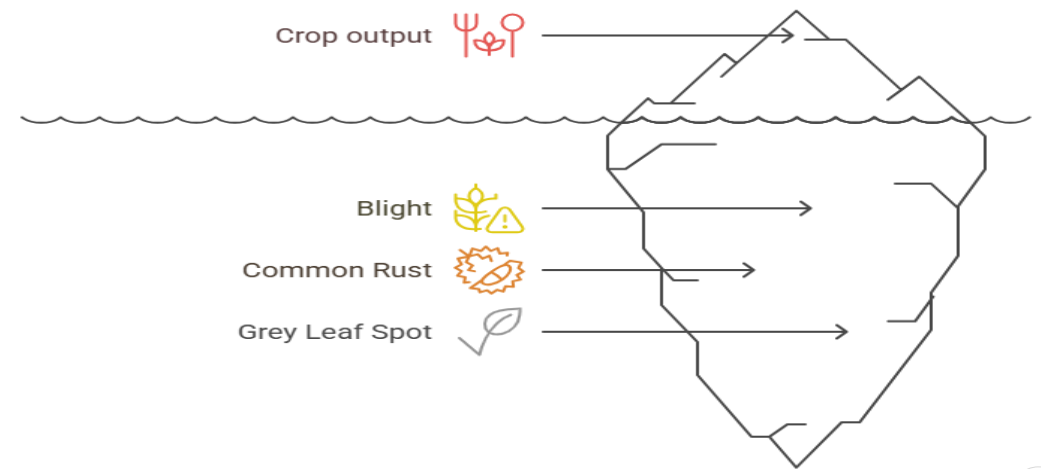
Introduction

Objective: Develop a mobile-based CNN system for early, real-time detection of maize leaf diseases.



- Maize (*Zea mays* L.) is a key cereal crop for food security, animal feed, and industrial uses.
- Manual inspection is slow, labor-intensive, and requires experts.

Foliar diseases threaten crop output.

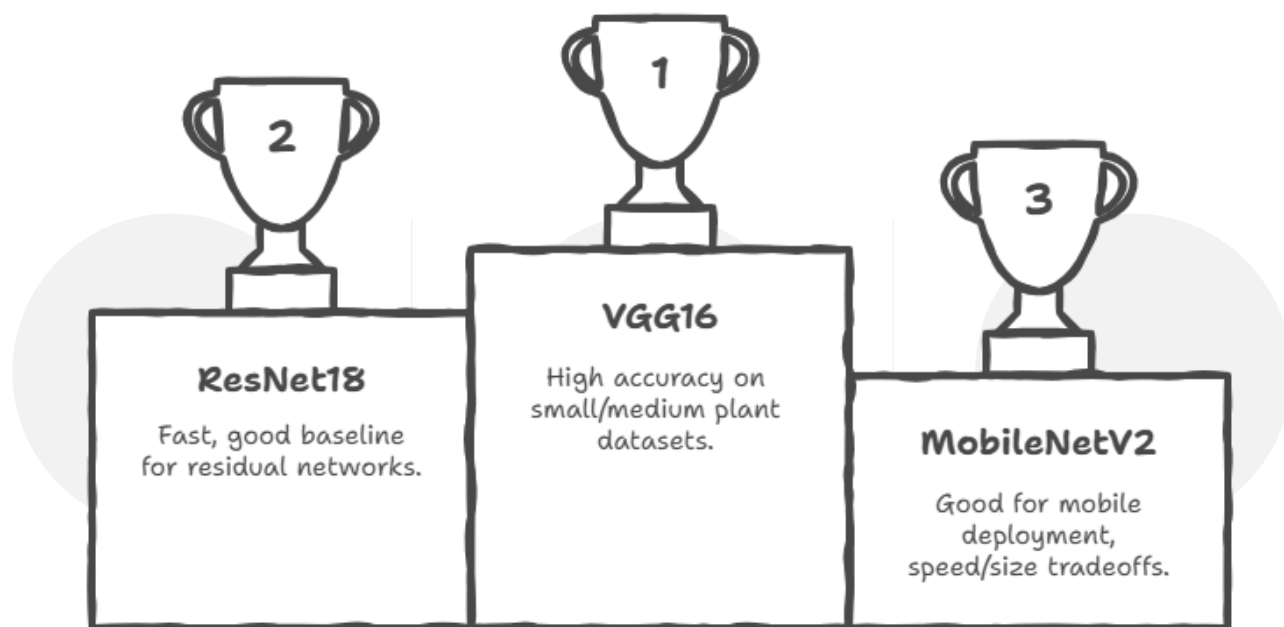




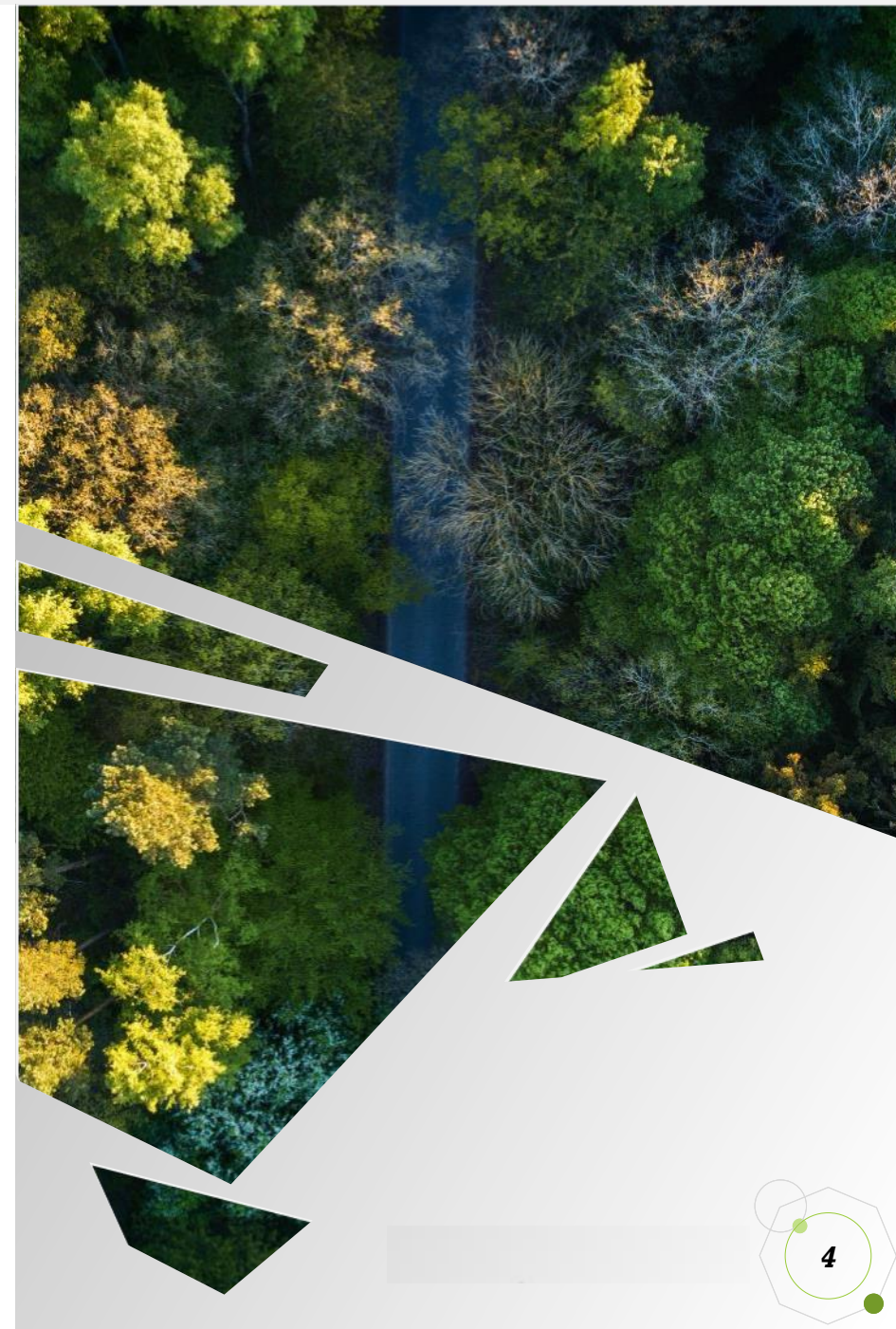
Dataset & Preprocessing

- **Dataset:** CV-Corn-Disease-Detection (GitHub, 2023)
 - 4 classes: Blight, Common Rust, Grey Leaf Spot, Healthy
- **Preprocessing and Augmentation:**
 - Resizing, normalization, and rotation to improve image quality
 - Balanced classes to reduce class imbalance
- Enables **robust training** for real-world detection

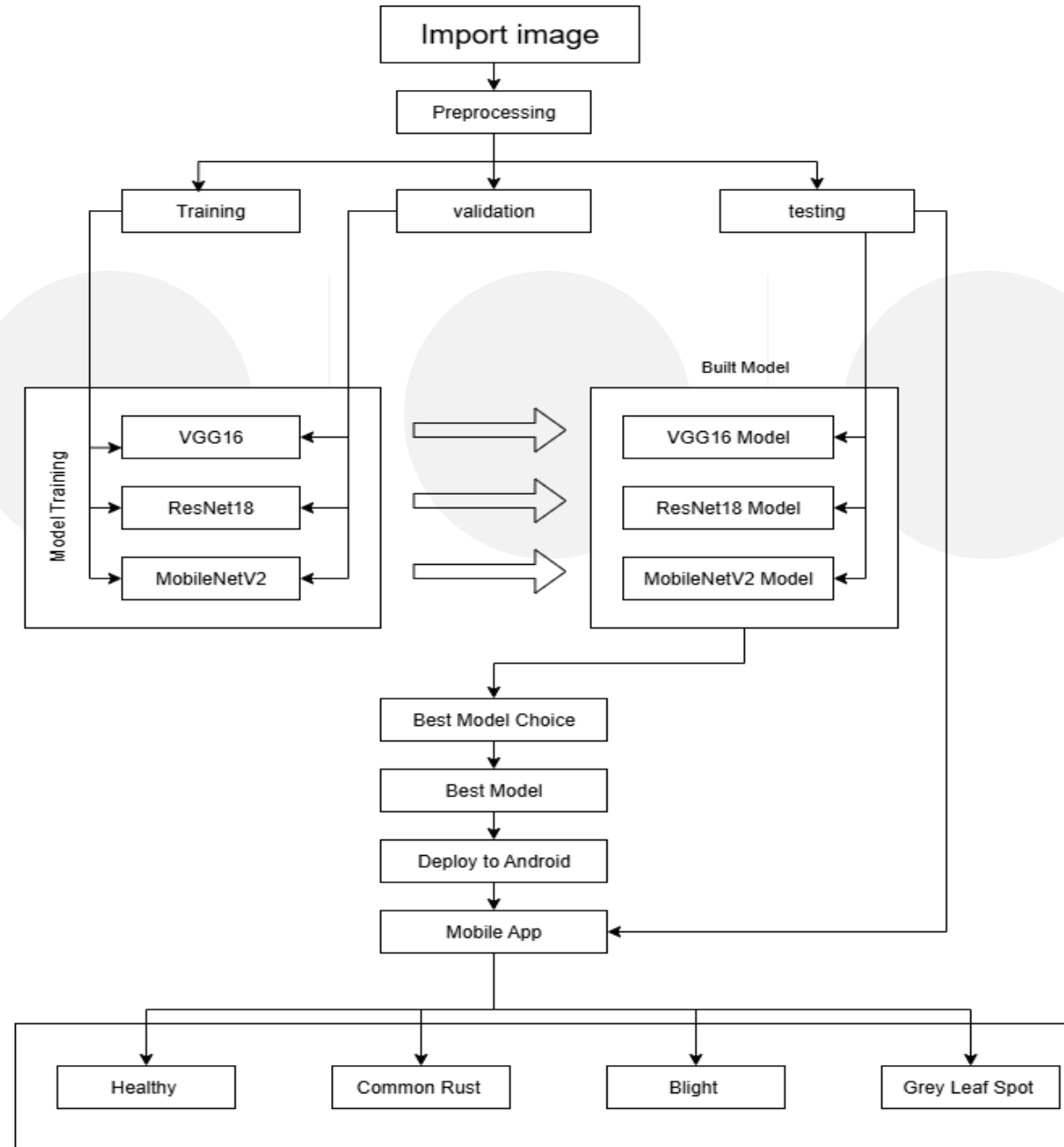
Top Plant Dataset Models

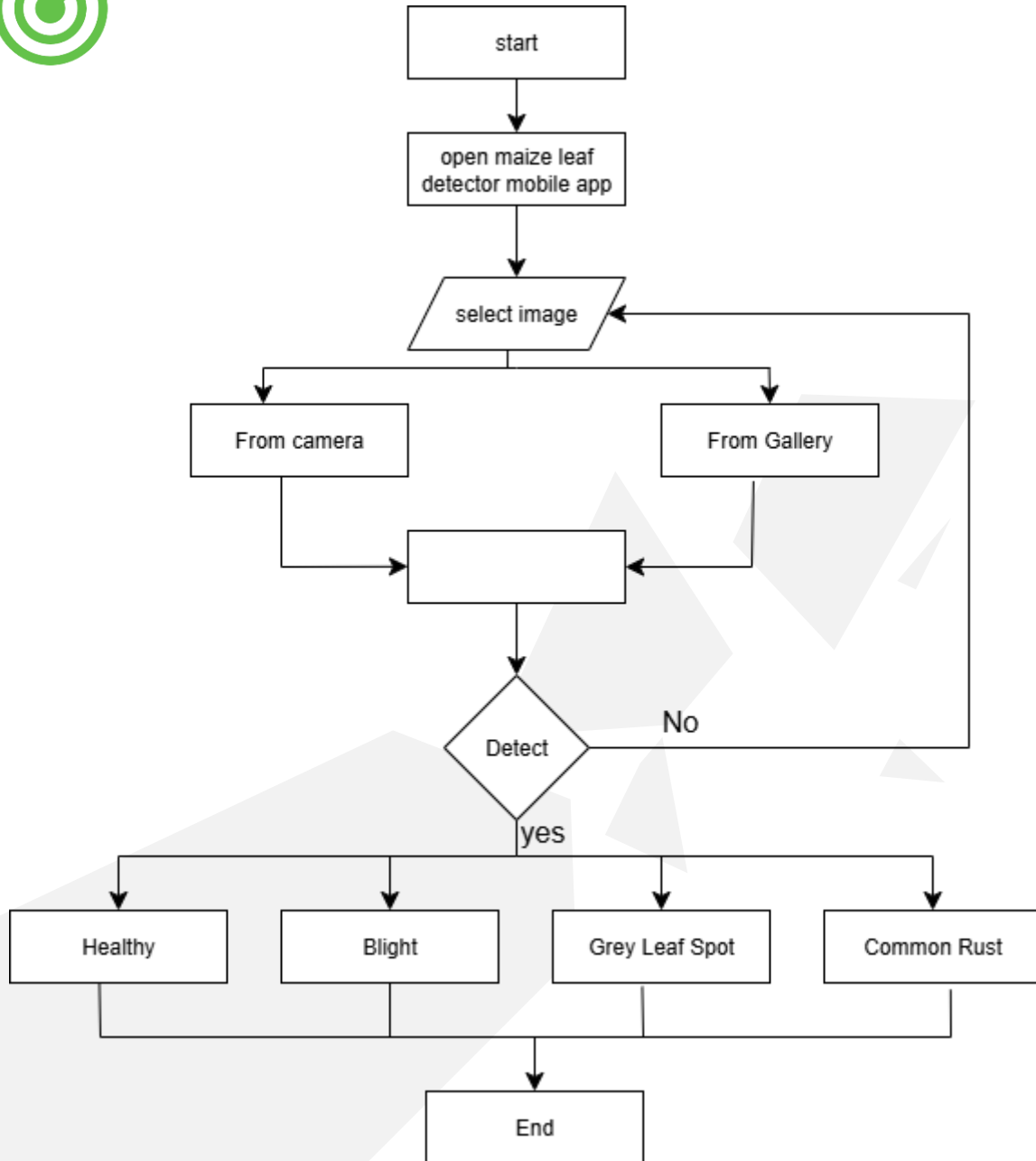


Model	TA	VA	TstA
VGG 16	0.62	0.73	0.76
ResNet18	0.99	0.94	0.92
MobileNetV2	0.99	0.92	0.87



Methodology





Corn Disease Detection

Is your green buddy dying?

Try to identify the cause and get extensive disease and care info in a snap.

Upload a photo of a corn leaf or plant and our model will identify common diseases — with actionable advice for farmers and agronomists.

Fast Analysis

Get a diagnosis in seconds (demo: mock model).

Field-ready Tips

Practical treatment & prevention suggestions for corn growers.



Personal plant doctor is with you.

Have you ever searched 'what's wrong with my plant'? The results may have been disappointing... No more with Us! Simply snap a photo of the issue to get a diagnosis. We will give you detailed info on the disease, what caused it, how to treat it, and how to prevent it.

Upload an image

Drag & drop or click to select a photo (leaf or ear close-ups work best).

Select photo

Analyze



Detection Result

Disease: Northern Corn Leaf Blight

Confidence: 92.0%

Recommended actions

Remove severely infected leaves, rotate crops, consider resistant hybrids.

Download report

Clear

An aerial photograph of a coastal area. On the left, there is a parking lot with several cars and a few small buildings. A road curves through a green, tree-filled area. The beach is sandy and stretches across the middle. To the right, the water is a clear, light blue-green, and a large pile of rocks is visible at the shoreline. The text "Thank You" is overlaid in the center in a white serif font on a dark rectangular background.

Thank You